

BITS, PILANI – K. K. BIRLA GOA CAMPUS

Theory of Computation

(CS F51)

Lecture No. 11



FSA & Regular Languages

Theorem:

The class of languages accepted by FSA is *closed* under

1. Union
2. Concatenation
3. Kleene Star
4. Complementation
5. Intersection

Once we prove 1 & 4, **5** follows from *De Morgan's Law*

We have *already seen* results 1 & 4 for DFA

Note:

We have proved similar result for *regular languages*

FSA & Regular Languages

Proof (for *union*):

Suppose L_1 & L_2 are languages with $L_1 = L(M_1)$ & $L_2 = L(M_2)$ where

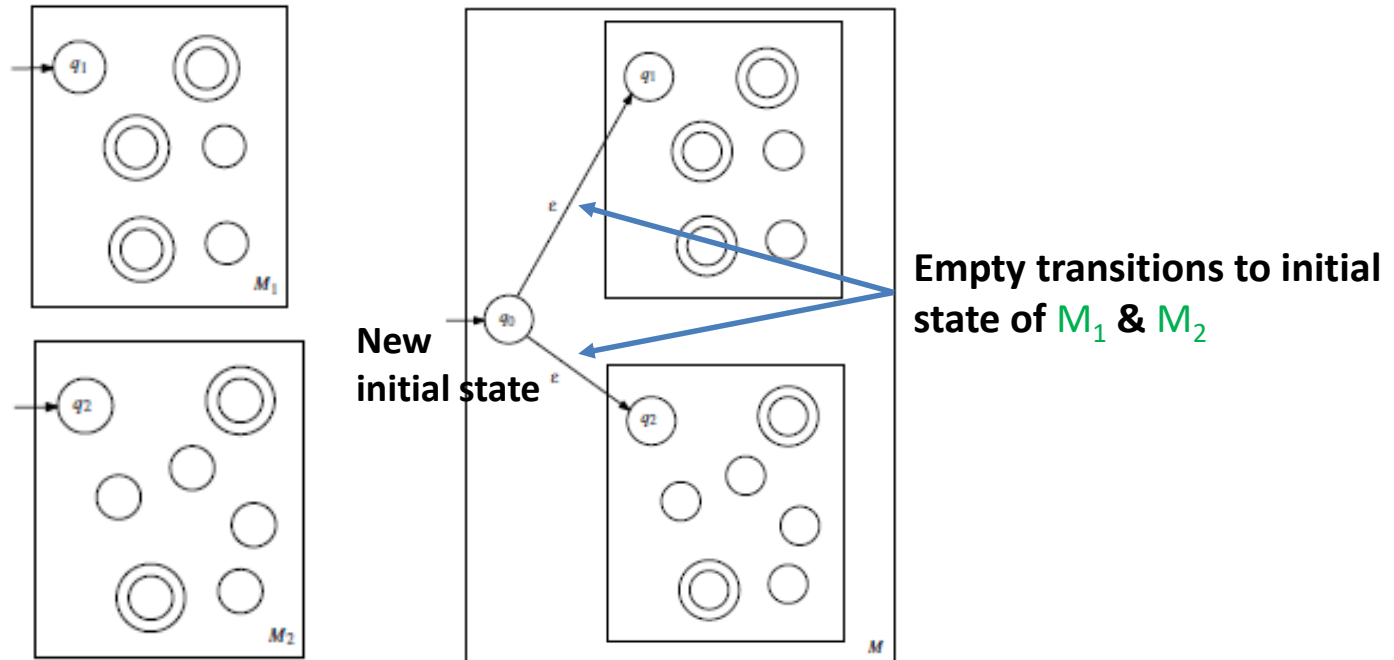
$M_1 = (K_1, \Sigma, \Delta_1, q_1, F_1)$ & Let $M_2 = (K_2, \Sigma, \Delta_2, q_2, F_2)$

We assume $K_1 \cap K_2 = \phi$, otherwise *rename* the states

We will construct $M = (K, \Sigma, \Delta, q_0, F)$ such that $L(M) = L(M_1) \cup L(M_2)$

We *imitate* the idea of the example:

M begins any computation by *nondeterministically* choosing to enter initial state of M_1 or M_2 & then *imitates* either M_1 or M_2



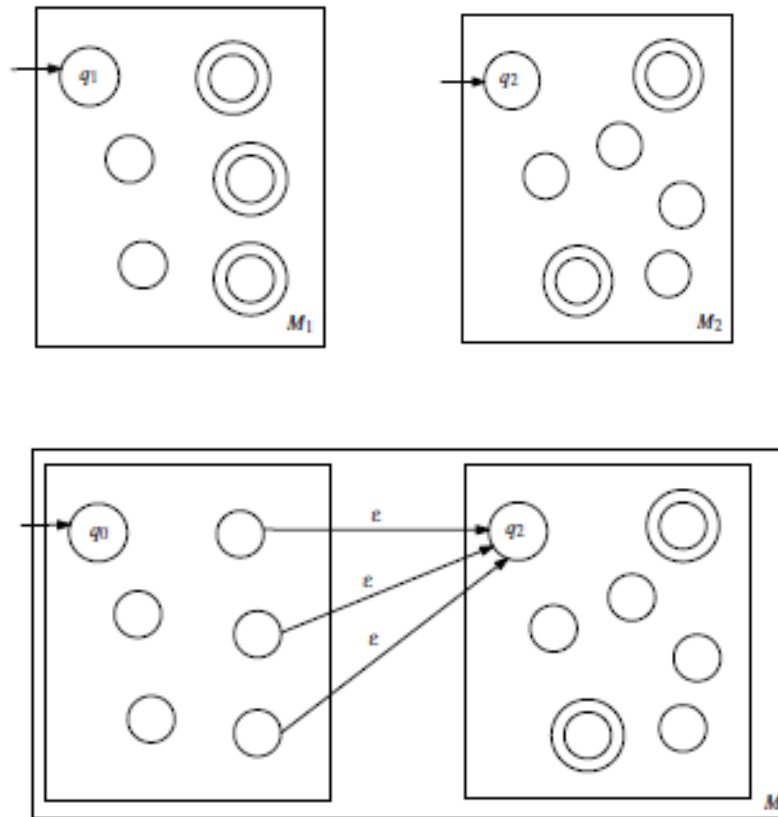
FSA & Regular Languages

For concatenation

We will construct $M = (K, \Sigma, \Delta, q_0, F)$ such that $L(M) = L(M_1) L(M_2)$

Intuitive idea

Start by *simulating* M_1 for a while & then *moving nondeterministically* (*guessing* the part from L_1) from a *final state* of M_1 to the *initial state* of M_2 & then *imitate* M_2



FSA & Regular Languages

For Kleene Star

We will construct $M = (K, \Sigma, \Delta, q_0, F)$ such that $L(M) = L(M_1)^*$

Intuitive Idea

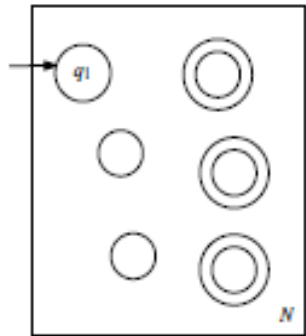
Imitating concatenation idea

For constructing M , In M_1 , From each of the final state **add empty transition** to the initial state

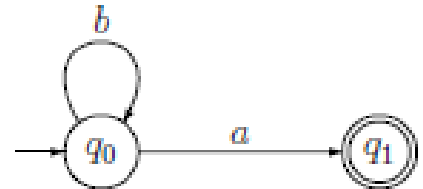
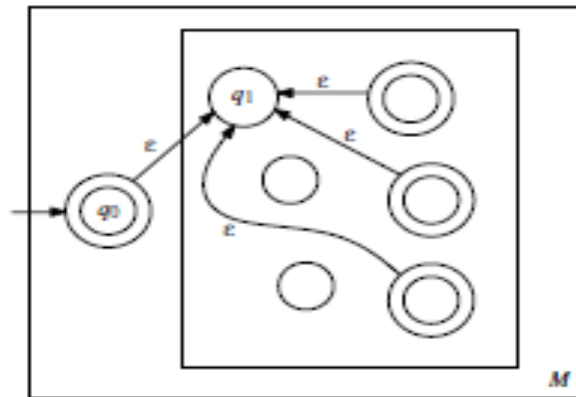
Problem1 – if empty string $\epsilon \notin L(M_1)$ then $\epsilon \notin L(M_1)^*$

Solution1 – Make initial state also a final state

Problem2 – If the initial state of M_1 is not accepting & has a **self loop**?



So we introduce **a new start state** q_0 (which **is final**), and it is connected by an **empty transition** to the initial state of M_1 .



Exercise

Complete the details of the proof

FSA & Regular Languages

Proof for *closure* under :

Complement

- *Swap* the final & the non-final states

Intersection

- *De Morgan's Law*

Main Result

Main Result

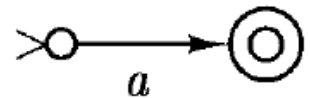
A language is regular if and only if it is accepted by a FSA.

Proof:

Remember class of *regular languages* is the *smallest class* of languages containing $\{a\}$, a is an alphabet and closed under union, concatenation & Kleene Star.

Observe all singletons say $\{a\}$ are accepted by FSA.

By theorem (proved just now), the class of languages accepted by FSA is *closed* under union, concatenation & Kleene Star.



$\therefore$ every regular language is accepted by some FSA.

Regular Expression to FSA

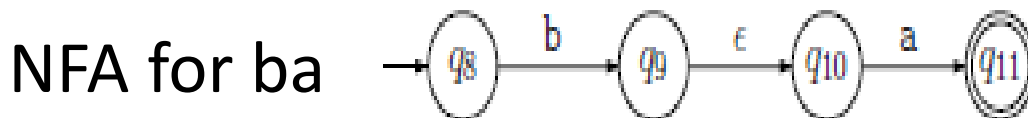
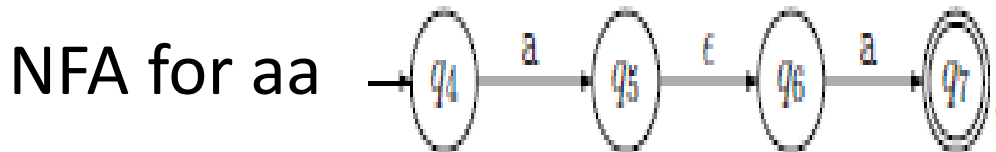
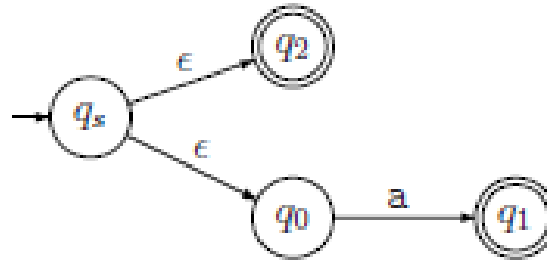
Note:

The above proof gives a *construction* to design an NFA accepting the given regular language

Consider the *regular expression* $(a \cup \varepsilon)(aa \cup ba)^*$

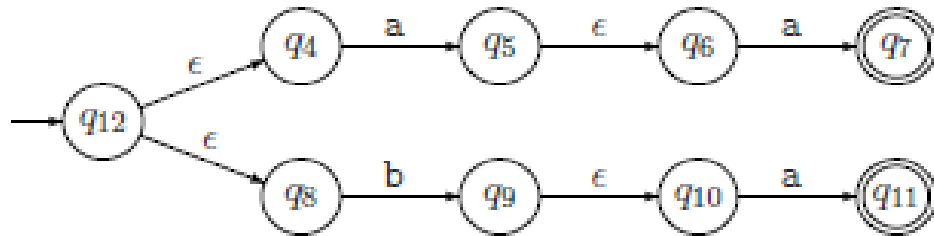
The NFA for ε is $\rightarrow q_2$ and for a is $\rightarrow q_0 \xrightarrow{a} q_1$.

The NFA for $(a \cup \varepsilon)$ is

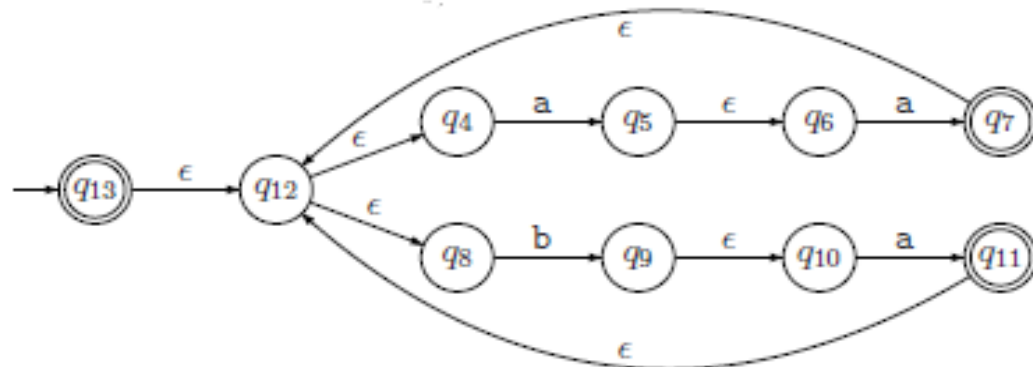


Regular Expression to FSA

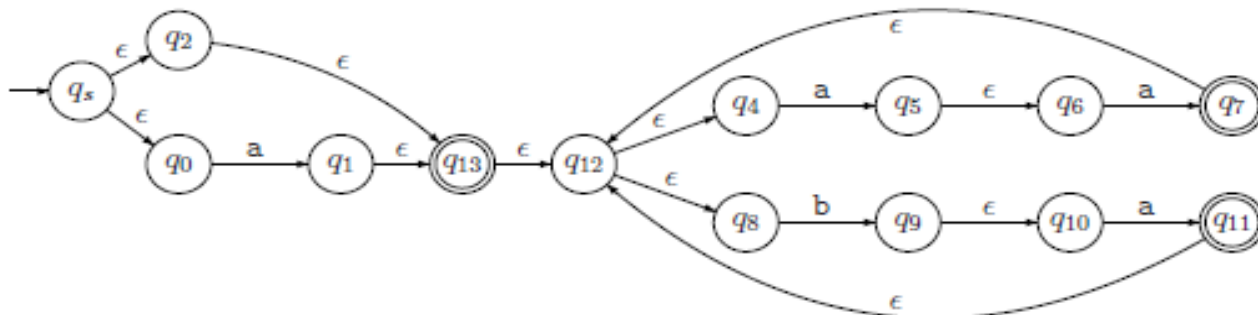
NFA for $aa \cup ba$



NFA for $(aa \cup ba)^*$



NFA for $(a \cup \varepsilon)(aa \cup ba)^*$



FSA to Regular Expression

Conversely, suppose $M = (K, \Sigma, \Delta, s, F)$ is an FSA

We shall construct a *regular expression* R such that $L(R) = L(M)$

Idea: We will show that $L(M)$ is the union of many simple languages each of which is *regular*.

Let $K = \{q_1, q_2, \dots, q_n\}$ & $s = q_1$

For $i, j = 1, 2, \dots, n$ & $k = 0, 1, \dots, n$ we define

$R(i, j, k)$ as the set of all strings in Σ^* that may drive M from state q_i to state q_j *without passing* through any intermediate state numbered $k+1$ or greater.

Note: q_i & q_j are allowed to be numbered higher than k

FSA to Regular Expression

Observe: When $k = n$, it follows that

$$R(i, j, n) = \{w \in \Sigma^* : (q_i, w) \vdash_M^* (q_j, e)\}$$

$$\therefore L(M) = \cup \{R(1, j, n) : q_j \in F\}$$

Claim: All of $R(i, j, k)$ are *regular* & hence so is $L(M)$.

Proof of the claim is by *induction* on k .

Base Case: $k = 0$

$$R(i, j, 0) = \{a \in \Sigma \cup \{e\} : (q_i, a, q_j) \in \Delta\}$$

Now, each of $R(i, j, 0)$ is finite & so *regular*.

Inductive Hypothesis: Assume $R(i, j, k-1)$ for all i, j are *regular*.

FSA to Regular Expression

Inductive Step: We will show that each set $R(i, j, k)$ can be defined combining previously defined *regular languages* & using the operations of union, concatenation & Kleene Star.

Observe: We can go from q_i to q_j without passing through a state numbered greater than k in either of the following ways:

(1) Go from q_i to q_j without passing through a state numbered greater than $k-1$ **OR**

(2) **a.** q_i to q_k then **b.** from q_k to q_k zero or more times **c.** then q_k to q_j and in each case without passing through a state numbered greater than $k-1$

Therefore

$$R(i, j, k) = R(i, j, k-1) \cup R(i, k, k-1) R(k, k, k-1)^* R(k, j, k-1)$$